

Sprint 2 — Business Case

CS 5388 / CS 4390 Software Project Management

Team: 1 **Project:** Damn Parking **Date:** October 5, 2026

1) Purpose and Business Need

Damn Parking addresses uncertainty about parking availability. Interviews documented drivers circling lots, waiting for departing vehicles, and choosing more distant spaces. One Phase 1 participant reported a 40-minute parking search and lateness to work. The intended benefit is better-informed parking decisions; reduced search time has not yet been demonstrated.

This business case concerns the Live Parking Monitoring slice, not a complete commercial deployment. It supports a team decision about a potential project. The course activity is estimation and planning; the estimated implementation hours are not hours already worked during Sprint 2.

2) Proposed Scope and Beneficiaries

The six-story slice includes nearby-lot lookup, live availability comparison, directions, a manager view of patterns and possible misuse, parking filters, and available-space counts. Drivers receive location and occupancy information; managers receive a summary for human review. The implementation assumption is an existing service that supplies occupancy observations, rather than building a new computer-vision model. Consistent with the Sprint 1 charter, the system should use occupancy information without collecting unnecessary identifying information. The manager view supports human review; it is not an enforcement or identity-tracking system.

Excluded work includes camera installation, new model development or training, payments, reservations, automated enforcement, and full commercial deployment. Earlier ideas for 10–15 lots and roughly 60 cameras with continuous coverage are future expansion options, not part of the estimated baseline. User departure reporting and predictions of when parking will become available in the broader Release 1 list are outside these six estimated stories. The Sprint 1 prototype remains the broader concept; this estimate does not demonstrate completion of its full success criteria.

3) Options for the Team to Evaluate

Option	Implication
Keep current search methods	Avoids new development spending; drivers continue using maps, direct observation, and informal information.
Develop the limited slice	Tests whether availability information can support driver and manager tasks within a bounded scope.
Pursue the broader camera launch	Requires a separate scope, data-access plan, infrastructure budget, and implementation estimate.

The recommendation is to continue validating the limited slice before committing to implementation or commercial deployment.

4) Cost and Schedule Evidence

Measure	Planning estimate
Effort range	340–680 team hours
Expected effort	538 team hours
Loaded labor rate	\$60/person-hour; classroom assumption
Labor cost range	\$20,400–\$40,800
Expected labor cost	\$32,280
Team capacity	4 people × 20 hours/week = 80 team hours/week
Elapsed effort-based schedule	4.25–8.5 weeks; expected 6.73 weeks

The range includes 40 team hours for integration and coordination. It excludes separately billed hosting, maps, camera access, ongoing support, and other operating costs. Calendar estimates assume capacity is available and exclude external waiting. The Estimation Appendix contains the calculations.

5) Value and Remaining Uncertainty

Potential value includes avoiding searches in full lots, comparing options before arrival, and providing occupancy information to managers. These benefits depend on data accuracy, freshness, coverage, and usability. The research does not establish willingness to pay, customer acquisition, recurring revenue, or quantified savings.

The ROI Analysis therefore provides break-even thresholds and illustrative scenarios, not a forecast. Relevant evidence to collect includes camera-service access terms, occupancy accuracy under representative conditions, operating-cost quotes, and feedback from prospective users and lot operators. The scope of “misuse” also needs definition; occupancy data alone does not establish every type of misuse.

6) Team Recommendation and Justification

The Project Manager evaluates readiness and presents the go/no-go recommendation. Significant decisions affecting scope, budget, major requirements, or project objectives require discussion and approval by the project team and, when applicable, the Project Sponsor. The Project Manager handles routine decisions within the approved scope and escalates decisions beyond that authority.